

HDOC Day-30 Activity

Menu-Driven C Program on Array Operations

Name	_____
Roll No.	_____
Batch	15 / 16
Subject	Problem Solving / C Programming

Problem Statement

Write a C program to perform the following menu-driven operations on a given array: (1) Sort the elements using Bubble Sort and perform a Binary Search, (2) perform insertion and deletion at a valid position, (3) rotate the array elements to the right by k positions, and (4) exit the program.

C Program

```
#include <stdio.h>

#define MAX 100

void display(int a[], int n)
{
    int i;
    printf("\nArray elements: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n");
}

void bubbleSort(int a[], int n)
{
    int i, j, temp;

    for (i = 0; i < n - 1; i++)
    {
        for (j = 0; j < n - i - 1; j++)
        {
            if (a[j] > a[j + 1])
            {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }
}

int binarySearch(int a[], int n, int key)
{
    int low = 0, high = n - 1, mid;

    while (low <= high)
    {
        mid = (low + high) / 2;

        if (a[mid] == key)
            return mid;

        if (a[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }

    return -1;
}

void insertElement(int a[], int *n)
{

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    int pos, value, i;

    if (*n >= MAX)
    {
        printf("\nArray is full.\n");
        return;
    }

    printf("Enter position (1 to %d): ", *n + 1);
    scanf("%d", &pos);

    if (pos < 1 || pos > *n + 1)
    {
        printf("Invalid position.\n");
        return;
    }

    printf("Enter element: ");
    scanf("%d", &value);

    for (i = *n; i >= pos; i--)
        a[i] = a[i - 1];

    a[pos - 1] = value;
    (*n)++;

    printf("Element inserted successfully.\n");
}

void deleteElement(int a[], int *n)
{
    int pos, i;

    if (*n == 0)
    {
        printf("\nArray is empty.\n");
        return;
    }

    printf("Enter position (1 to %d): ", *n);
    scanf("%d", &pos);

    if (pos < 1 || pos > *n)
    {
        printf("Invalid position.\n");
        return;
    }

    for (i = pos - 1; i < *n - 1; i++)
        a[i] = a[i + 1];

    (*n)--;

    printf("Element deleted successfully.\n");
}

void rotateRight(int a[], int n)
{
    int k, i, j, temp;

    if (n == 0)
    {
        printf("\nArray is empty.\n");
        return;
    }

    printf("Enter number of positions to rotate right: ");
    scanf("%d", &k);

    if (k < 0)
    {
        printf("Invalid number of positions.\n");
        return;
    }

    k = k % n;

    for (i = 0; i < k; i++)
    {
        temp = a[n - 1];

        for (j = n - 1; j > 0; j--)

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        a[j] = a[j - 1];

        a[0] = temp;
    }

    printf("Array rotated successfully.\n");
}

int main()
{
    int a[MAX], n, i, choice, key, result;

    printf("Enter number of elements (1 to %d): ", MAX);
    scanf("%d", &n);

    if (n < 1 || n > MAX)
    {
        printf("Invalid array size.\n");
        return 0;
    }

    printf("Enter %d array elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Bubble Sort and Binary Search\n");
        printf("2. Insertion and Deletion\n");
        printf("3. Rotate Array Right\n");
        printf("4. Exit\n");
        printf("===== \n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                bubbleSort(a, n);
                printf("\nArray after Bubble Sort:");
                display(a, n);

                printf("Enter element to search: ");
                scanf("%d", &key);

                result = binarySearch(a, n, key);

                if (result != -1)
                    printf("Element found at position %d.\n", result + 1);
                else
                    printf("Element not found.\n");
                break;

            case 2:
                printf("\n--- Insertion ---\n");
                insertElement(a, &n);
                display(a, n);

                printf("\n--- Deletion ---\n");
                deleteElement(a, &n);
                display(a, n);
                break;

            case 3:
                rotateRight(a, n);
                display(a, n);
                break;

            case 4:
                printf("\nProgram exited successfully.\n");
                break;

            default:
                printf("\nInvalid choice. Please try again.\n");
        }
    } while (choice != 4);

    return 0;
}

```

Result

The menu-driven C program performs Bubble Sort with Binary Search, insertion, deletion, right rotation, and exit operations successfully on an array.